

Bonsai | Project progress

30 September 2026 | Model-in-Silicon: Bonsai 2 27B ternary target

Foundation delivered; first executable slice validated in simulation. Week 0 is complete, Week 1 is active, and Weeks 2-6 remain backlog. Custom Bonsai FPGA results and full-model token generation are not yet demonstrated.

EVIDENCE DELIVERED

Week 0 / PR #1 merged. Interface/weight-store contracts, image tooling and environment are delivered. The existing baseline passed **1,332 selected targets**; vendor XSIM passed **9/9**. The official F2 example executed **2,264 additions** on hardware, validating the environment rather than Bonsai inference.

Week 1 / PR #2 draft. The sealed DDR/store -> DOT128 -> mailbox path is implemented. Local validation passed **16 steps: 396 native + 6 synthetic cases** (two expected invalid-code rejections); both full-image readbacks matched. The transport passed **41 SDK mock checks** and CodeQL passed on 3ee0f99e. These are software/simulator results.

MILESTONE SCHEDULE

Milestone	Status	Completion gate / planned outcome
Week 0 - Baseline and interfaces	DONE	PR #1 merged; report and actual vendor FPGA evidence
Week 1 - Core RTL and tooling	ACTIVE	Custom XSIM passed; Placement/routing active; AFI/physical gates pending
Week 2 - Ternary engine and fit	BACKLOG	Correctness, synthesis and measured FPGA fit
Week 3 - Helpers and intrinsics	BACKLOG	Integrated helpers, kernels and layer-level tests
Week 4 - Timing and F2 fit	BACKLOG	Named configurations with routed timing/resources
Week 5 - Attention and eight tiles	BACKLOG	Attention, eight-engine feasibility and inference candidate
Week 6 - Scheduling and sign-off	BACKLOG	Scheduling, end-to-end evaluation and release evidence

Week 1 planned due: 2 October 2026. This is the issue's planned date, not a completion forecast. Week labels express the milestone sequence.

NEXT THREE DELIVERABLES

- 1. Complete implementation and physical FPGA gates:** XSIM on 839e5fc1 **passed 87 assertions: 3 actual Q1_0 cases + 1 synthetic case**. All 4,160 fixture bytes, exact dots and raw FP16 scales matched (test finish: 64.224 simulated us; exit 0). Three failed attempts are retained. Synthesis completed; placement/routing is active; no routed result, AFI or full 242 MB physical run yet. The real SDK transport built on F2 (exit 0), without hardware execution.
- 2. Attach the current evidence report:** publish this project snapshot and refresh the Week 1 report with exact source, simulator and actual FPGA artifacts as they become available.
- 3. Freeze the HBM array implementation contract:** the HBM design and independent review are published; no array RTL is implemented. Next freeze interfaces, numerics and ownership. Preferred FPGA storage is **HBM: load, verify, then seal as ROM-equivalent**. HBM is volatile; implemented DDR remains the baseline, and ASIC ROM is future work.

PRINCIPAL RISKS AND BOUNDARIES

The current **Bonsai 1.7B Q1_0 fixture is binary sign quantization**; it does not validate the separate **27B ternary** target. Custom timing closure, memory bandwidth and eight physical engines remain unproven. Full Arcilator support and model integration remain follow-on work. Accountable owner: **Rachit Tibrewal**; Codex implements and runs checks with separate automated technical review. Each milestone closes with a reviewed/merged PR, runnable simulator, actual FPGA evidence for its scope and an attached report.

SOURCES

[Week 0 evidence](#) | [Week 1 local evidence](#) | [XSIM results and limits](#) | [HBM design](#) | [HBM review](#) | [Project plan](#)

30 September snapshot: custom XSIM passed; placement/routing active, no routed result. No full-model tokens, physical throughput or Arcilator cycles/second measured.